Sabarish Venkataramani Malathi

Education

2022-2026 **PhD in Physics**, *University of Hamburg*, Germany

2020-2022 **Master of Science in Physics (Cosmology Track)**, *RWTH Aachen University*, Germany, CGPA - 1.3.

2017-2019 **Master of Science in Physics**, *University of Madras*, Chennai, Tamil Nadu, India, CGPA - 8.016/10.0.

2013-2016 **Bachelor of Science in Physics**, *RKM Vivekananda College, University of Madras*, Chennai, Tamil Nadu, India, CGPA - 9.06/10.0 - **First Rank Holder**.

Publications

4 refereed scientific publications since 2022, 2 as first author and another two as co-first author. A complete list available on NASA/ADS.
1 refereed pedogogical publication meant for high-school students.

Scientific

2025 **Sabarish, V.M.**, Brüggen, M., Schmidt-Hoberg, K. & Fischer, M. S., *A&A 703, A142*

2024 **Sabarish, V.M.**, Brüggen, M., Schmidt-Hoberg, K., Fischer, M. S. & Kahlhoefer, F., *MNRAS 529, 2032–2046*

2024 Casas, S., Lesgourgues, J., Schöneberg, N., **Sabarish V.M.** , et al., *A&A 682, A90*.

2025 Euclid Collaboration et. al, *A&A 693, A58 (2025)*.

Pedagogical

2020 **Sabarish V.M.**, Apparent Weight of a Photon Box. *Resonance* 25, 911–914

Mentoring Experience

Master thesis

2024-2025 Supervised a master thesis in Computational astrophysics.

Teaching Experience

Tutor at University of Hamburg

The courses are tailored to beginning undergrads. Tutor responsible for grading and discussing homework for the following courses.

2024 Astronomy and astrophysics I

2024 Astronomy and astrophysics II

Teaching assistant at the University of Hamburg : Python for physics

Summary Delivering both lectures and discussing homework solutions. I also contributed to designing the course.

Private high school tutor (2017)

Summary I was tutoring high school students part-time during my university studies. I was mostly tutoring physics, but sometimes due the absence of other tutors, I had to cover Mathematics and Computer Science as well.

Computer skills

Languages Python(advanced), C(intermediate), C++(intermediate) ,Fortran(basic), Rust(intermediate), Bash(advanced)

Others Mathematica, LateX, git

Coding projects in Physics

All source files for these projects can be found at github.com/sabarish-vm

Unitconvert Python and a mathematica package that converts between any unit system(natural, Plank, Gaussian, SI, etc.)

Geodesics A Mathematica notebook that can animate orbits (3D and 2D) of a test mass around a black hole.

zTox A command line tool to quickly convert redshift to distances

Experience

Talks

2024 Conference : IDM 2024, *L'Aquila, Italy*

2023 Workshop : Self-Interacting Dark Matter: Models, Simulations and Signals, *Pollica, Italy*

2023 Parallel session : Quantum Universe day, *Hamburg*

2025 Parallel session : Quantum Universe day, *Hamburg*

School

2024 NBIA Summer school on Astrophysical dynamics of gravitating systems, *Niels Bohr Institute, Copenhagen, Denmark*

2022 Summer school on Modified Gravities, Inflation, and Cosmological perturbations, *CSIC, Barcelona, Spain*

2022 Code-Astro workshop on Programming for Astronomy, *Caltech, USA*

Achievements and Leadership Positions

2022- Co-organizer of SIDM journal club

2019 Student Coordinator for the Workshop "Cosmology : A New Addition to the Precision Physics"

2019 Seetharaman Endowment award in recognition of performance in MSc